

Solution Requirements

Date	04 july2026
Team ID	XXXXXX
Project Name	Opticrop-Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Farmers and administrators can securely log in using username, password, or mobile number verification.</i>	High
2	Authorization levels	<i>Farmers can access crop recommendations and reports. Admins can manage users, datasets, and system settings.</i>	High
3	External interfaces	<i>Connects with weather APIs, soil databases, and agricultural datasets to fetch real-time information.</i>	High

4	Transactions processing	<i>Collects user inputs (soil type, rainfall, temperature, location) and processes them through ML algorithms to generate crop recommendations.</i>	High
5	Reporting	<i>Generates crop recommendation reports, yield prediction reports, and visual dashboards showing crop suitability.</i>	Medium
6	Business rules, etc.	<i>Recommendations are based on soil nutrients, climate conditions, rainfall, and historical agricultural data. Only valid agricultural data is accepted.</i>	High
7	Compliance to laws or regulations	<i>Ensures user data privacy and follows agricultural data management guidelines and applicable data protection regulations.</i>	Medium
8	<i>Other</i>	<i>Provides multilingual support and crop cultivation guidance for recommended crops.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>System should quickly analyze input data and provide recommendations.</i>	<i>Recommendation generated within 2 seconds</i>
2	Scalability	<i>System should support increasing numbers of farmers and agricultural datasets.</i>	<i>Supports 5,000+ concurrent users</i>
3	Security & Data Privacy	<i>Protect user information and agricultural data through secure storage and communication.</i>	<i>AES-256 encryption and secure authentication</i>
4	Reliability & Availability	<i>System should remain operational and accessible throughout the farming season.</i>	<i>99.9% uptime per month</i>
5	Usability & Accessibility	<i>Interface should be simple for farmers and accessible on mobile and desktop devices.</i>	<i>User-friendly UI with responsive design</i>
6	<i>Other</i>	<i>System should be easy to update and deploy on different platforms.</i>	Modular architecture and cross-platform support Project Details